Model Line Coverage Report

Group 44

This report gives the line coverage that the JUnit tests achieve on the system model, as required by the project brief. The model is the `model` package and its sub-packages.

1 Result

The tests execute **512 of 529 lines (96.8%)** of the model code. All 35 classes and 125 of 126 methods are executed. All 35 tests pass.

Package	Classes	Methods	Lines
`model`	4/4 (100%)	19/19 (100%)	98/101 (97%)
`model.criteria`	14/14 (100%)	38/38 (100%)	171/171 (100%)
`model.exceptions`	5/5 (100%)	9/10 (90%)	9/10 (90%)
`model.persistence`	3/3 (100%)	17/17 (100%)	130/139 (93%)
`model.records`	9/9 (100%)	42/42 (100%)	104/108 (96%)
Total	35/35 (100%)	125/126 (99%)	**512/529 (96.8%)**

Branch coverage of the model is 275 of 303 branches (90%). Figure 1 is the IntelliJ IDEA coverage window for this run.



Figure 1: Coverage window of IntelliJ IDEA after running all tests.

2 How the figures were measured

- **Tools:** IntelliJ IDEA with its built-in coverage runner, on OpenJDK 24.0.2 (language level 17), JUnit 4.13.1.
- **Tests run:** all classes in `src/test/java/tests`: `RecordTests`, `CriteriaTests`, `StorageTests` (28 model tests) and `ConsoleTests` (7 console scenarios). The result was 35 passed, 0 failed.
- **Scope:** only the `model` packages are counted. The `controller`, `view` and `app` packages are not part of this report.
- **To repeat:** open the PIM folder in IntelliJ IDEA, right-click `src/test/java/tests`, choose *Run with Coverage*, and read the *Line* column of the coverage window.

3 Uncovered lines

The 17 lines that the tests do not reach are spread over four packages: 3 in `model`, 1 in `model.exceptions`, 9 in `model.persistence` and 4 in `model.records`. No line of `model.criteria` is missed.